

Plants, Part 2: Photosynthesis

Quick reference sheet · Biology · keep handy for revision

1. Where photosynthesis happens

Photosynthesis takes place inside **chloroplasts**, which contain **chlorophyll** — a green pigment that absorbs light energy. **Palisade mesophyll cells**, near the top of the leaf, are packed with chloroplasts and adapted to absorb the most light.

2. The photosynthesis equation

carbon dioxide + water → glucose + oxygen

(light energy, absorbed by chlorophyll, in the chloroplasts)

Photosynthesis is **endothermic** — it takes in light energy rather than releasing it. CO₂ diffuses in through the stomata; water arrives via the xylem. Glucose can be used for respiration, stored as starch, or used to make cellulose or proteins. Oxygen is released as a waste product through the stomata.

3. Factors affecting the rate of photosynthesis

Factor	Increases the rate when...	Why
Light intensity	higher	more light energy for chlorophyll to absorb
CO ₂ concentration	higher	more raw material available for the reaction
Temperature	higher, up to a point	reactions speed up; too hot denatures enzymes, slowing it back down

A **limiting factor** is whichever factor is in shortest supply, stopping the rate rising further even if the other two increase — shown as the point a rate graph levels off.

4. Translocation

Glucose made in the leaf is often converted to **sucrose** and transported around the plant (to growing parts or storage organs) through the **phloem**, in both directions. This is called **translocation**.

5. Plant nutrition & disease

Deficiency	Symptom	Why
Nitrate	Stunted growth	needed to make proteins, for cell growth
Magnesium	Yellow leaves (chlorosis)	needed to make chlorophyll

Plants can also be affected by disease (fungi, bacteria, viruses) or insect attack — look for stunted growth, leaf spots, decay, growths, discolouration or wilting, and compare against reference charts to identify the cause.